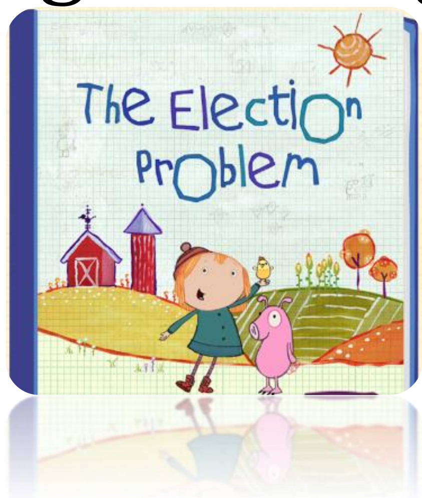
# CS 170 Programming Quiz 4




For PQ04, you'll complete an IPO console program from scratch. For this exam you'll need to know:

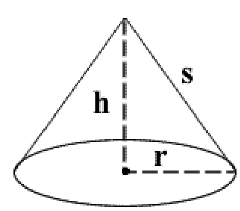
- How to do input and output with `Scanner` and `System.out`
- How to read a formula and convert it to a Java expression.
- How real division and integer division differ
- How to use the `Math` class functions and constants
- How to import the correct packages
- How to write the program in the correct IPO order.

This exam will be tested in two parts. Part I will check to see that the format exactly matches the mockup that is provided. Part II will check that your calculations are correct.

Below are some sample problems that you can use for practice.

## 1 THE CONE VOLUME PROBLEM

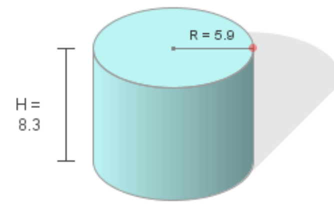
Write a program to calculate the **volume** of a right-circular code, like the one pictured here. To calculate the volume (**V**) of a cone, given a radius (**R**) and a height (**H**), the formula is:




$$V = \frac{1}{3}\pi R^2 H$$

## 2 THE CYLINDER AREA PROBLEM

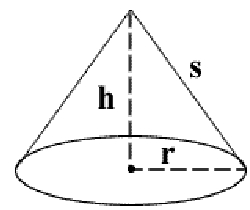
Write a program to calculate the **surface area** of a cylinder; you'll need the area of the top, plus the area of the bottom, plus the area around the cylinder (the lateral surface area). The formula for calculating the total surface area of a cylinder (**A**) with radius (**R**) and height (**H**) is:




$$A = 2\pi R^2 + 2\pi RH$$

# 3 The Cone Surface Area Problem

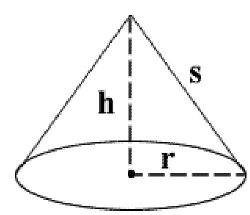
Write a program to calculate the **total surface area** of a right-circular code, like the one pictured here. To calculate the area (**A**) of a cone, given a radius (**r**) and a height (**h**), the formula is:




$$A = \pi r(r + \sqrt{h^2 + r^2})$$

# 4 The Cone Slant Height Problem

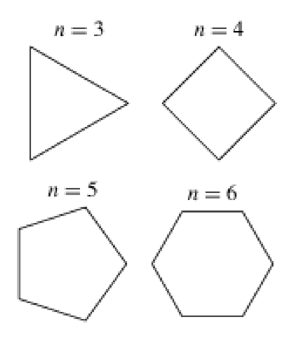
Write a program to calculate the **slant height** (**S**) of a right-circular code, like the one pictured here. Given a radius (**r**) and a height (**h**), the formula is:




$$S = \sqrt{h^2 + r^2})$$

# 5 The Polygon Area Problem

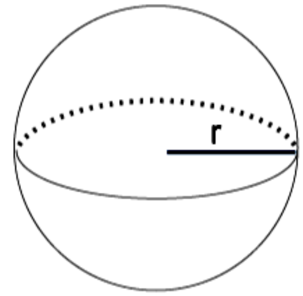
Write a program to calculate the **surface area** of a regular polygon, one where all sides have the same length, like those in the picture to the right. To calculate the surface area you need the number of sides (**n**) and the length of any side (**s**). The formula you'll use is:




$$area = \frac{s^2 n}{4\tan\left(\frac{\pi}{n}\right)}$$

# 6 THE SPHERE STATISTICS PROBLEM

Write a program to calculate both the volume and the area of a sphere, given a radius **r**. The volume **V**, and the area **A**, are calculated by using the formulas:

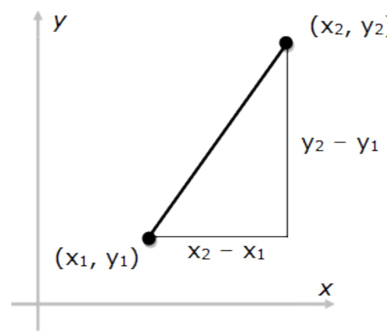



- $V = \frac{4}{3}\pi r^3$
- $A = 4\pi r^2$

# 7 THE POINT DISTANCE PROBLEM

Write a program to calculate the distance between two points (**x1**, **y1**) and (**x2**, **y2**) and displays the distance. The formula for calculating the distance between two points is:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



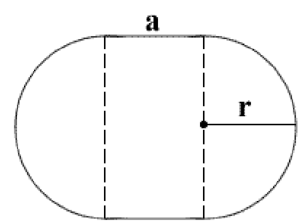

# 8 THE NEW WINDCHILL PROBLEM

Write a program to calculate the "wind-chill" or "feels-like" temperature (**T**) given the wind speed (**v**) in miles per hour and the temperature (**t**) in Fahrenheit. Your program should use the "new" wind-chill formula adopted in 2001 by the National Weather Service:

$$T = 35.74 + 0.6215t - 35.75v^{0.16} + 0.4275tv^{0.16}$$

## 9 THE STADIUM AREA PROBLEM

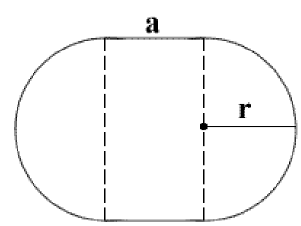
A stadium is a circle of radius **r** that has been cut in half through the center and the 2 ends are then separated by a rectangle of height **r** and width (or side length) of **a**. Write a program to calculate the area (**A**) using this formula.




$$A = \pi r^2 + 2ra$$

## 10THE STADIUM PERIMETER PROBLEM

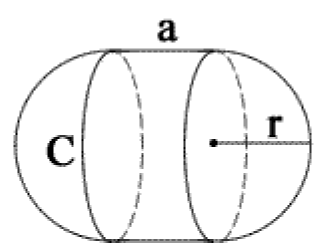
A stadium is a circle of radius **r** that has been cut in half through the center and the 2 ends are then separated by a rectangle of height **r** and width (or side length) of **a**. Write a program to calculate the perimeter (P) using this formula.




$$P = 2(\pi r + a)$$

## 11 THE CAPSULE VOLUME PROBLEM

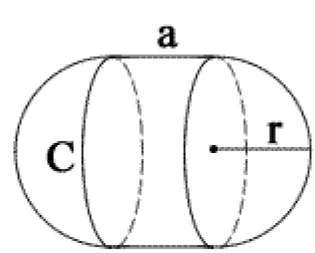
Write a program to calculate the volume (**V**) of a capsule with a radius **r** and a side length **a**, using the formula:




$$V = \pi r^2 \left( \frac{4}{3} r + a \right)$$

## 12 THE CAPSULE AREA PROBLEM

Write a program to calculate the surface area (**A**) of a capsule with a radius **r** and a side length **a**, using the formula:

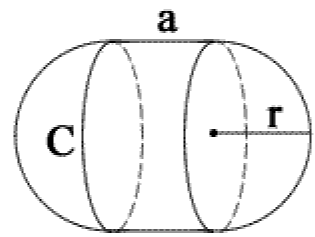



$$V = 2\pi r(2r + a)$$

## 13 THE CAPSULE CIRCUMFERENCE PROBLEM

Write a program to calculate the circumference (**C**) of a capsule with a radius **r** and a side length **a**, using the formula:

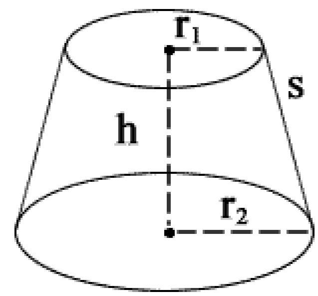
$$C = 2\pi r$$




## 14 THE CONICAL FRUSTUM VOLUME PROBLEM

Write a program to calculate the volume (**V**) of a conical frustum with a height **h** and the two radius $\mathbf{r_1}$ and $\mathbf{r_2}$. Here is the formula:

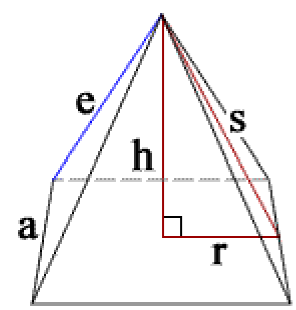
$$V = \frac{1}{3}\pi h\left(r_1^{\,2} + r_2^{\,2} + (r_1 r_2)\right)$$




## 15 THE PYRAMID VOLUME PROBLEM

Write a program to calculate the volume (**V**) of a square pyramid as shown here. Here is the formula:

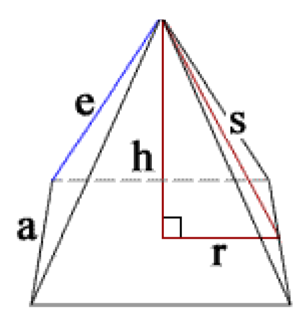
$$V = \frac{1}{3}a^2 h$$




## 16 THE PYRAMID SURFACE AREA PROBLEM

Write a program to calculate the total surface area (**A**) of a square pyramid as shown here. Here is the formula:

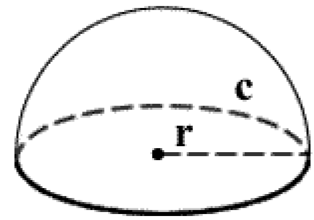
$$A = a\left(a + \sqrt{a^2 + 4h^2}\right)$$



## 17 THE HEMISPHERE VOLUME PROBLEM

Write a program to calculate the volume (**V**) of a hemisphere as shown here. Here is the formula:

$$V = \frac{2}{3}\pi r^3$$




## 18 THE HEMISPHERE SURFACE AREA PROBLEM

Write a program to calculate the total surface area (**K**) of a hemisphere as shown here. Here is the formula:

$$K = 3\pi r^2$$